

Industrial processes are increasingly using enzymes to promote uncommon or stereospecific chemical reactions that would be difficult to achieve with conventional chemical catalysts and reaction conditions. However, a common challenge in these biocatalysis methods is preserving the stability of the enzyme once it has been purified. Directed evolution, a technique used to find mutants with increased thermostability, can be performed via error-prone PCR or by replicating the protein-encoding DNA in organisms with defective mismatch repair systems. Increased thermostability is often correlated with increased rigidity of the protein (through additional stabilizing noncovalent bonds or covalent disulfide bridges) and often comes at a cost of decreased catalytic efficiency.

An alternative approach is to start with enzymes from thermophilic organisms, whose proteins have evolved to be extremely stable even at high temperatures. At moderate temperatures, these enzymes typically demonstrate poor catalytic efficiency, so directed evolution is performed to increase catalytic efficiency at the cost of decreased stability.

A particular enzyme (Enzyme **X**) from *Thermoanaerobacter Brockii* was subjected to this approach. Enzyme **X** catalyzes the reversible conversion of acetophenone to 1-phenylethanol (Figure 1). Researchers found that a combination of A85G and I86A mutations in the enzyme binding pocket resulted in large changes to Enzyme **X**'s Michaelis–Menten parameters (Table 1).

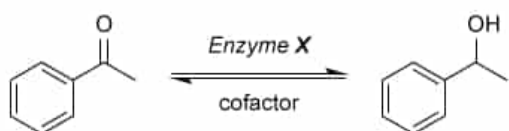


Figure 1 Reaction of acetophenone by Enzyme **X**

Table 1 Michaelis–Menten Parameters of Enzyme **X** Before and After Directed Evolution Measured at 30 °C

Mutation	K_m (mM)	k_{cat} (min ⁻¹)
Wild-type	19.0 ± 2.7	1.80 ± 0.27
A85G/I86A	3.97 ± 0.47	102.4 ± 6.1

Interestingly, the researchers found that the A85G/I86A mutation alters the stereoselectivity of the enzyme. Whereas the wild-type protein is only poorly enantioselective, producing a slight excess of the (*S*)-enantiomer, the mutant produces almost exclusively the (*R*)-product. Researchers repeated their kinetic characterization of wild-type and mutant Enzyme **X** by examining the effects on different ketone-containing substrates and found similar changes to catalytic efficiency and stereoselectivity.

What class of enzyme is Enzyme **X**?

✔ ☒ A. Oxidoreductase [87%]

☐ B. Hydrolase [6%]

☐ C. Lyase [1%]

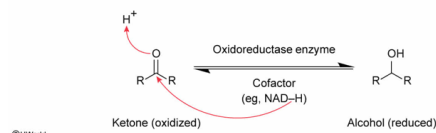
☐ D. Isomerase [4%]

Correct

86% answered correctly

Explanation:

Oxidoreductase enzymes catalyze the reduction of ketones to alcohols



Enzymes are broadly **classified** into categories based on the type of reaction they catalyze.

1. **Oxidoreductases** are enzymes that catalyze **oxidation-reduction** reactions of their substrates using **cofactors** as either reducing agents (eg, NADH, FADH_2 , NADPH) or oxidizing agents (eg, NAD^+ , FAD, NADP^+).
2. **Transferases** transfer functional groups from **one molecule to another**. Transferases receive **two substrates** and produce **two products**.
3. **Hydrolases** catalyze hydrolysis reactions, which use **water** to break functional groups apart (eg, **phosphatase**, gluconolactonase) (**Choice B**).
4. **Lyases** are enzymes that either break a **single substrate into two products** or join two substrates to make a single product, often with the breaking or formation of **double bonds** or a **ring structure**. Lyase reactions occur **without** the use of water, redox cofactors, or a separate high-energy molecule like ATP (**Choice C**).
5. **Isomerases** rearrange covalent bonds within **one molecule**. Isomerase-catalyzed reactions can include functional group transfers (mutases), stereochemical rearrangement (ie, racemases, epimerases), *cis-trans* rearrangement, and conformational rotation of macromolecules (eg, chaperones) (**Choice D**).
6. **Ligases** (eg, DNA ligase, ubiquitin ligase) use the hydrolysis of a separate, **high-energy molecule** (eg, ATP) to provide the free energy change to join two substrates together. Note that ligases *also* have ATP hydrolase activity, but their primary activity is to join two substrates.

The passage and Figure 1 indicate that Enzyme **X** catalyzes the conversion of acetophenone to 1-phenylethanol with the help of a cofactor. In other words, a **ketone** functional group is converted to an **alcohol**, which has a **lower oxidation state** (ie, a **reduction**). A cofactor (eg, NADH) was needed to serve as a reducing agent. Therefore, **Enzyme X is classified as an oxidoreductase**.

Educational objective:

Enzymes are classified by the type of reaction they catalyze. An enzyme that catalyzes the reduction of a ketone to an alcohol with the help of a cofactor is an oxidoreductase.

Time spent: 0 sec

Subject:
Biochemistry

QID: 403897

System:
Enzymes

Industrial processes are increasingly using enzymes to promote uncommon or stereospecific chemical reactions that would be difficult to achieve with conventional chemical catalysts and reaction conditions. However, a common challenge in these biocatalysis methods is preserving the stability of the enzyme once it has been purified. Directed evolution, a technique used to find mutants with increased thermostability, can be performed via error-prone PCR or by replicating the protein-encoding DNA in organisms with defective mismatch repair systems. Increased thermostability is often correlated with increased rigidity of the protein (through additional stabilizing noncovalent bonds or covalent disulfide bridges) and often comes at a cost of decreased catalytic efficiency.

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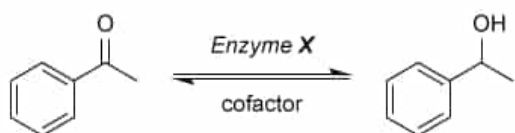


Figure 1 Reaction of acetophenone by Enzyme **X**

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Presumably, the enzymes display different rates at different temperatures because:

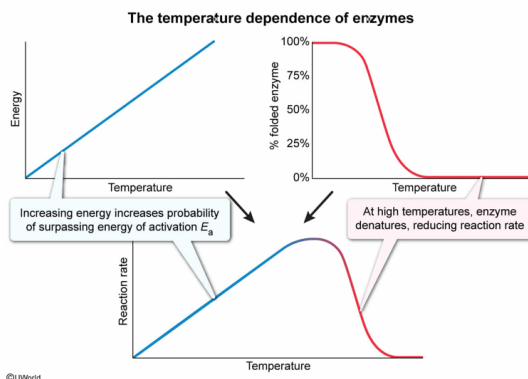
- I. the average energy of the enzyme's substrates increases with increasing temperature.
- II. the probability of the enzyme's denaturing increases with increasing temperature.
- III. the enzyme's effect on the ΔG of the reaction increases with increasing temperature.

- ☐ A. II only [24%]
- ✓ ☒ B. I and II only [55%]
- ☐ C. II and III only [10%]
- ☐ D. I, II, and III [9%]

Correct

56% answered correctly

Explanation:



Enzymes are biological catalysts that **speed up** chemical reactions, allowing them to occur under the mild conditions within living organisms. The enzymes in organisms have evolved through natural selection to have an **optimal temperature** at which the enzyme catalyzes its reaction at the fastest rate. As temperature varies from this optimal temperature, the reaction rate decreases.

Enzymes speed up reactions by lowering the **activation energy** E_a of a reaction. Although E_a is lowered relative to the uncatalyzed reaction, E_a continues to act as an **energy barrier** and represents the **energy needed by the substrates** to result in a productive **collision**. **Increasing the temperature** of the reaction increases the fraction of substrate molecules with **enough energy** to surpass the lowered activation energy. Therefore, as the temperature increases to the optimal temperature, the rate of reaction also increases (**Number I**).

However, raising the temperature too high can have a detrimental effect on the rate of the catalyzed reaction. As temperatures rise, the **noncovalent bonds** involved in maintaining **secondary, tertiary, and quaternary structure** begin to break, causing the enzyme to **denature**. Denatured enzymes are no longer functional and cannot catalyze reactions, so the reaction rate decreases at very high temperatures (**Number II**).

(**Number III**) Although the change in Gibbs free energy (ΔG) of a reaction often changes with temperature, enzymes change only E_a while leaving ΔG *unaffected*. Enzymes have no effect on ΔG , regardless of temperature.

Educational objective:

Enzymes are biological catalysts that speed up chemical reactions to allow them to occur within living organisms. Enzymes have an optimal temperature for catalysis at which the rate of the enzyme-catalyzed reaction is greatest. This temperature achieves a balance: high enough for substrates to surpass the activation energy but low enough to maintain a folded, functional enzyme.

Time spent:0 sec

Subject:
Biochemistry

QID:403898

System:
Enzymes

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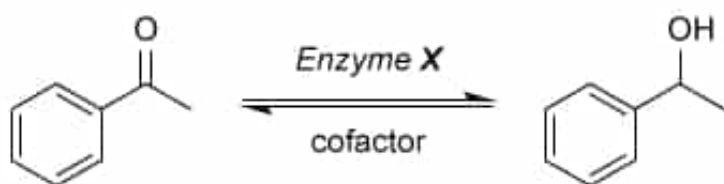


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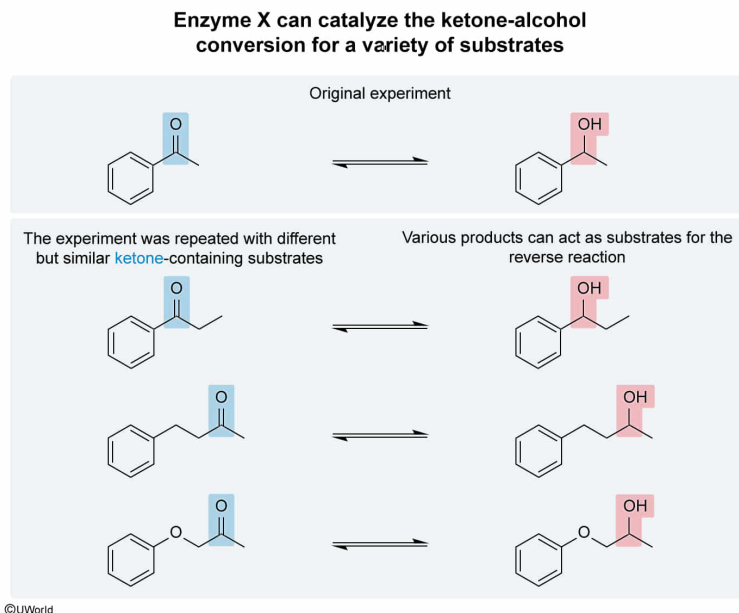
The information in the passage best supports which conclusion about the specificity of Enzyme **X**?

- ☐ A. Wild-type Enzyme X is highly specific for a single alcoholic substrate in the reverse reaction. [9%]
- ☐ B. Wild-type and mutant Enzyme X can use similar primary, secondary, or tertiary alcohols as substrates for the reverse reaction. [4%]
- ☐ C. Mutant Enzyme X is highly specific for a single ketone-containing substrate in the forward reaction. [47%]
- ✓ ☒ D. Wild-type Enzyme X can act on a variety of structurally similar substrates both in the forward reaction and in the reverse reaction. [39%]

Correct

39% answered correctly

Explanation:



Different enzymes have **different levels of specificity** toward their substrates. Some enzymes are **highly specific** and will catalyze reactions only with a **single molecule**. Other enzymes are **less specific** and will catalyze reactions with **several different molecules** that have **related structures**. The specificity of an enzyme can be determined by observing its ability to act on related molecules.

The passage describes Enzyme **X**, which catalyzes the **reversible** conversion of the ketone-containing molecule acetophenone into the hydroxyl-containing 1-phenylethanol (ie, the reduction of the ketone). The passage further states that the kinetics for wild-type and mutant Enzyme **X** were originally characterized with acetophenone but that the **experiment was repeated** with **different ketone-containing substrates**. These repeated experiments showed **similar changes to catalytic efficiency**, indicating that Enzyme X was able to successfully use a **variety of related molecules as substrates**. As a reversible reaction, the variety of products also allow for the use of multiple substrates in the reverse reaction.

(Choice A) The passage states Enzyme **X** is able to reduce multiple related ketones, which would create multiple related

alcohols to act as substrates for the reverse reaction. Moreover, the passage also indicates that wild-type Enzyme **X** produces both enantiomer products, with only a *slight* excess of the (S)-enantiomer. Therefore, at least two related alcoholic molecules (ie, the (R)-enantiomer and the (S)-enantiomer) can act as substrates for the reverse reaction.

(Choice B) The reverse of the described reaction is the oxidation of a secondary alcohol to a ketone. Although Enzyme **X** might be able to oxidize primary alcohols, *tertiary* alcohols **cannot be oxidized**.

(Choice C) The passage states that mutant Enzyme **X** is able to reduce multiple related ketones for the forward reaction, as indicated by successful measurement and detection of change in the mutant's catalytic efficiency.

Educational objective:

Some enzymes are promiscuous and able to catalyze reactions with a variety of structurally related substrates.

Time spent:0 sec

Subject:
Biochemistry

QID:403899

System:
Enzymes

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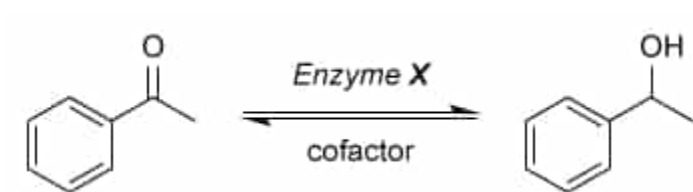


Figure 1 Reaction of acetophenone by Enzyme **X**

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According to Table 1, the catalytic efficiency for the A85G/I86A mutant at 30 °C is approximately:

- ☐ A. 250 times less than that of the wild-type enzyme. [5%]
- ☐ B. 2.5 times less than that of the wild-type enzyme. [2%]
- ☐ C. 2.5 times greater than that of the wild-type enzyme. [10%]
- ✔ ☒ D. 250 times greater than that of the wild-type enzyme. [81%]

Correct

81% answered correctly

Explanation:

The A85G/I86A mutant has 250× greater catalytic efficiency than wild type

Mutation	K_m (mM)	k_{cat} (min^{-1})	Catalytic efficiency (k_{cat} / K_m) ($\text{min}^{-1} \cdot \text{mM}^{-1}$)
Wild-type	19.0 ± 2.7	1.80 ± 0.27	$\approx \frac{2}{20} = 0.1$
A85G/I86A	3.97 ± 0.47	102.4 ± 6.1	$\approx \frac{100}{4} = 25$

$$\text{Relative change in catalytic efficiency} = \frac{\text{cat eff}_{\text{mutant}}}{\text{cat eff}_{\text{wild-type}}} = \frac{25 \text{ min}^{-1} \cdot \text{mM}^{-1}}{0.1 \text{ min}^{-1} \cdot \text{mM}^{-1}} = 250$$

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The Michaelis–Menten kinetic parameters of an enzyme are used to describe the ability of an enzyme to catalyze a chemical reaction. The **Michaelis constant** K_m describes the affinity of the enzyme–substrate interaction and is the concentration of free substrate needed for half the available enzyme active sites to be bound to substrate. The parameter

$V_{\max}$ is an **extensive property** that describes the maximum amount of product per unit time (ie, under saturating substrate conditions) that a **given amount of enzyme** can produce. The **turnover number** k_{cat} is the **intensive** analogue to $V_{\max}$ (ie, k_{cat} does not depend on the amount of enzyme) and describes the number of product molecules produced *per enzyme molecule*.

The intensive parameters of K_m and k_{cat} can be combined into a single value, the **catalytic efficiency**:

$$\text{cat eff} = \frac{k_{\text{cat}}}{K_m}$$

Catalytic efficiency is directly proportional to both k_{cat} and $V_{\max}$ (as either k_{cat} or $V_{\max}$ increase, catalytic efficiency increases) and is inversely proportional to K_m (the smaller the K_m , the stronger the affinity and the greater the catalytic efficiency). Different enzymes and their mutants can be compared with regard to their catalytic efficiencies.

Based on Table 1, the catalytic efficiency for the mutant can be approximated as:

$$\frac{k_{\text{cat,mutant}}}{K_{m,\text{mutant}}} = \frac{102.4 \text{ min}^{-1}}{3.97 \text{ mM}} \approx \frac{100 \text{ min}^{-1}}{4 \text{ mM}} = 25 \text{ min}^{-1} \cdot \text{mM}^{-1}$$

The catalytic efficiency for the wild-type enzyme can be approximated as:

$$\frac{k_{\text{cat,wild-type}}}{K_{m,\text{wild-type}}} = \frac{1.80 \text{ min}^{-1}}{19.0 \text{ mM}} \approx \frac{2 \text{ min}^{-1}}{20 \text{ mM}} = 0.1 \text{ min}^{-1} \cdot \text{mM}^{-1}$$

The relative catalytic efficiency of the mutant compared to the wild-type can then be calculated:

$$\frac{\text{cat eff}_{\text{mutant}}}{\text{cat eff}_{\text{wild-type}}} \approx \frac{25 \text{ min}^{-1} \cdot \text{mM}^{-1}}{0.1 \text{ min}^{-1} \cdot \text{mM}^{-1}} = 250$$

Therefore, the catalytic efficiency of the mutant is approximately **250 times greater** than that of the wild-type enzyme.

(Choice A) 250 times less would be calculated by comparing the wild-type enzyme to the mutant (ie, the mutant value is the reference value placed in the denominator). However, the question asks for the mutant to be compared to the wild-type.

(Choices B and C) A factor of 2.5 might be calculated if the k_{cat} were *multiplied* by the K_{m} rather than *divided* by it.

Educational objective:

Catalytic efficiency is the ratio of an enzyme's turnover number k_{cat} to its Michaelis constant K_{m} . The catalytic efficiencies of two enzymes can be compared by dividing the value of the compared enzyme by the value of the reference enzyme.

Time spent:0 sec

Subject:
Biochemistry

QID:403900

System:
Enzymes

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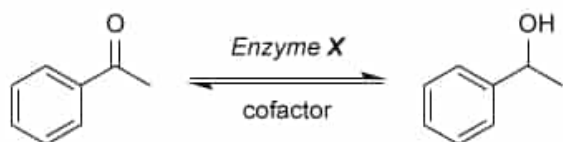


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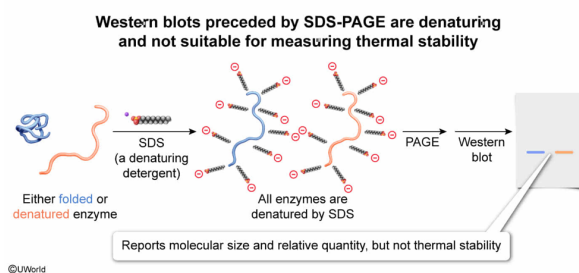
Which experimental method is LEAST suited to measuring the thermal stability of Enzyme X?

- ✓ ☒ A. Performing an SDS-PAGE followed by western blot against Enzyme X at different temperatures [55%]
- ☐ B. Measuring the heat capacity of Enzyme X at different temperatures [15%]
- ☐ C. Measuring circular dichroism (CD) spectra of Enzyme X at different temperatures [25%]
- ☐ D. Performing activity assays on Enzyme X at different temperatures [3%]

Correct

56% answered correctly

Explanation:



The **thermal stability** of an enzyme or protein refers to the ability of the protein to **resist denaturation at high temperatures**. Thermal stability of enzymes can be assessed in several ways.

- **The heat capacity** of an object is the amount of heat energy required to increase its temperature by 1 degree Celsius. Heat capacity **increases during phase transitions** (eg, at the **melting temperature of water**) because the energy is used to **break noncovalent intermolecular forces** rather than increase the average kinetic energy (ie, temperature).

Similarly, macromolecules like proteins and nucleic acids have a **melting temperature** T_m where the **heat capacity increases** as energy is used to break the **noncovalent bonds** that maintain their **structure**. Due to the **cooperative** nature of macromolecular folding, these bonds tend to break together, yielding a single T_m (**Choice B**).

- **Spectroscopic measurements** that report on protein conformation, such as **circular dichroism**, can also be used to report thermostability. Circular dichroism reports the relative amount and type of **secondary structure** within a protein, which is expected to change from primarily **α -helices and β -sheets** in folded proteins to primarily unstructured **random coils** in denatured proteins as the temperature rises. The T_m is the temperature at the halfway point of this transition (**Choice C**).

- **Activity assays** are experiments that measure the **function** of an enzyme. One way of doing this is by measuring the reaction rate. Enzymes are only functional if they are **folded correctly**, and therefore activity assays performed at different temperatures can also be used to assess the thermal stability of an enzyme **(Choice D)**.

The question asks which technique is **not suitable for measuring thermal stability**. The "SDS" in SDS-PAGE stands for **sodium dodecyl sulfate**, an anionic detergent that **denatures** proteins. The proteins run on the **PAGE (polyacrylamide gel electrophoresis)** gel are **always denatured**, regardless of temperature. Therefore, the subsequent **western blot** can be used to specifically visualize a desired protein to report its **size** and **relative amount** but not its thermal stability.

Educational objective:

Western blots following SDS-PAGE can be used to report molecular weight and relative quantities of proteins but not their thermal stability. In contrast, heat capacity measurements, spectroscopy, and activity assays at different temperatures can be used to measure thermal stability.

Time spent:0 sec

Subject:
Biochemistry

QID:403901

System:
Enzymes

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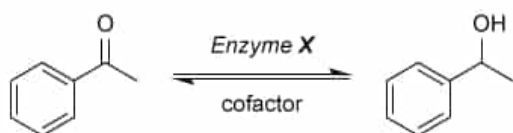


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Which of the following is the most likely explanation for the switch in enantioselectivity?

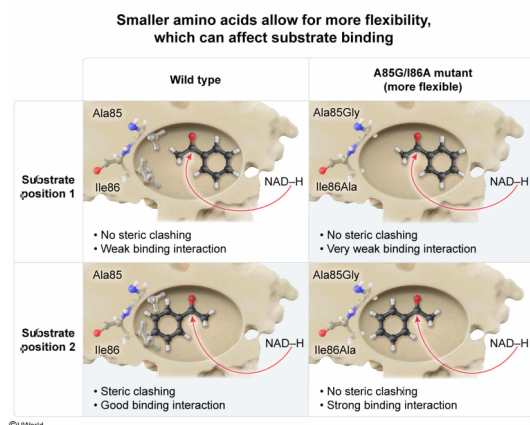
- ☐ A. The mutant side chains are more rigid and inflexible than those of the wild-type enzyme, forcing the substrate into a different binding pocket. [24%]
- ☐ B. The mutant side chains participate in an altered hydrogen bond network with the substrate compared to bonding in the wild-type enzyme, resulting in the substrate binding in the opposite orientation. [26%]

- ✓ ☒ C. The mutant side chains are smaller and more flexible than those of the wild-type enzyme, allowing the substrate to enter a previously occluded binding pocket. [44%]
- ☐ D. The mutant side chains have a charge opposite that of the wild-type enzyme, resulting in the substrate binding in the opposite orientation. [4%]

Correct

44% answered correctly

Explanation:



The specificity of enzymes for their substrates can be described by the **lock-and-key** and the **induced-fit models** for substrate binding. The lock-and-key model states that the shape of an enzyme's **empty** binding pocket perfectly complements the substrate so that only the correct substrate fits. In contrast, the induced fit model allows for ligand- and substrate-induced **conformational changes** after the small molecule and the enzyme interact, so that the complementarity between the empty binding pocket and the substrate does not need to be perfect.

Both models together can be used to explain aspects of the change in **enantioselectivity**. Two mutations were introduced into the enzyme-binding pocket (ie, the "lock"): A85G and I86A. In both cases, the original residues were mutated to **smaller, more flexible** amino acids. These changes likely allowed the substrate (ie, the "key") to access a differently shaped pocket. Specifically, the **bulky phenyl group** of acetophenone likely was able to access spaces that were previously **sterically inaccessible**, allowing the ketone to bind in an **orientation opposite** that of the ketone on the wild-type enzyme.

The induced fit model then accounts for conformational flexibility; the mutant enzyme can still bind strongly and catalyze the reduction reaction despite altered residues and opposite orientation of the substrate. Because **nucleophilic addition** to carbonyls (eg, by a hydride during a reduction reaction) produces stereoisomers based on which side of the carbonyl is attacked, the substrate's **opposite orientation** within the binding pocket due to **smaller, more flexible residues** leads to the **switch in enantioselectivity**.

(Choice A) Both mutations (alanine to glycine, and isoleucine to alanine) are to *smaller, more flexible* residues and not to bulkier, more rigid ones.

(Choice B) **Hydrogen bonds** require the hydrogens involved to be covalently linked to an *electronegative atom* (eg, N, O, F). In both the wild-type and the mutant side chains, all the hydrogen atoms are covalently linked to carbon, and therefore they will *not* participate in hydrogen bonding.

(Choice D) None of the mutant or wild-type residues discussed in the passage are charged residues.

Educational objective:

Glycine and alanine are smaller than alanine and isoleucine, respectively. Changes in the enzyme binding pocket alter the "lock" of the enzyme binding site, allowing substrates to bind in different orientations and reactions, which can affect reaction stereoselectivity.

Time spent:0 sec

Subject:
Biochemistry

QID:403902

System:
Enzymes